Seth Ricks ECEN 470 HW8: 6.3(a), 6.5(a), 6.6(b), 6.7(a), 6.8(a) 6.8(c), 6.11

HW8

(6.3) Sketch the asymptotes of the Bode plot magnitude and phase for the open-loop TF. Verify with MATLAB

a.) $L(s) = \frac{2000}{s(s+200)}$

$L(s) = \frac{2000}{s(s+200)} = \frac{2000}{s(200+s)}$ ** Canonical form: $L(s) = \frac{K\left(1+\frac{s}{\omega_{z1}}\right)\left(1+\frac{s}{\omega_{z2}}\right)\cdots}{(s)^?\left(1+\frac{s}{\omega_{p1}}\right)\left(1+\frac{s}{\omega_{p2}}\right)\cdots}$

$= \frac{\frac{1}{200}}{\frac{1}{200}} \cdot \frac{2000}{s(200+s)} = \frac{10}{s\left(1+\frac{s}{200}\right)}$

$L(j\omega) = \frac{10}{j\omega\left(1+\frac{j\omega}{200}\right)}$

** log power rule: $\log_b(M^K) = K\log_b(M)$

$|L(j\omega)|_{dB} = 20\log_{10}(|L(j\omega)|)$

$= 20\left(\log 10 - \left(\log(|j\omega|) - \log\left(\left|1+j\frac{\omega}{200}\right|\right)\right)\right)$

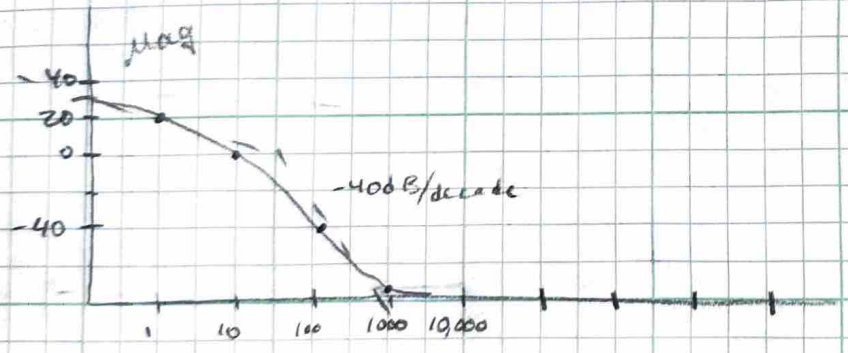
$= 20\log 10 - \left(\log(\omega) + \log\left(\sqrt{1+\left(\frac{\omega}{200}\right)^2}\right)\right)$

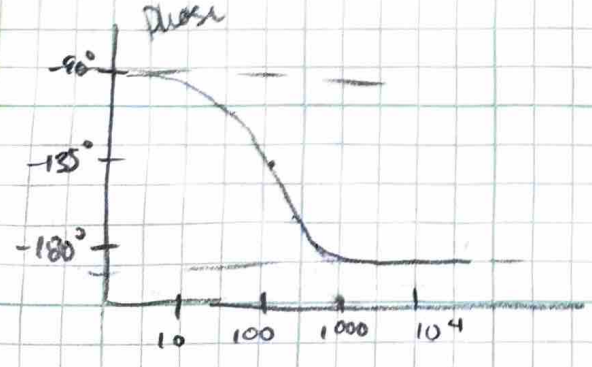
$= 20\log 10 - 20\log\omega - 20\log\sqrt{1+\frac{\omega^2}{40{,}000}}$

$= 20 - 20\log\omega - \frac{20}{2}\log\left(1+\frac{\omega^2}{40{,}000}\right)$

$= 20 - 20\log\omega - 10\log\left(1+\frac{\omega^2}{40{,}000}\right) =$

$\begin{cases} 20 - 20\log\omega, & \omega << 200 \\ -29.03, & \omega = 200 \\ 66 - 40\log\omega, & \omega >> 200 \end{cases}$





$L(j\omega) = \frac{2000}{j\omega(j\omega+200)} = \frac{10}{j\omega\left(1+j\frac{\omega}{200}\right)}$

$\angle L(j\omega) = \angle(10) - \angle j\omega - \angle\left(1+j\frac{\omega}{200}\right)$

$= 0° - 90° - \begin{cases} 0°, & \omega << 200 \\ 45°, & \omega = 200 \\ 90°, & \omega >> 200 \end{cases}$

=

(6.5) Sketch the asymptotes of the Bode plot for each of the listed open-loop TF, approximate transition at the second-order break-point, based on the value of the damping ratio. Verify with MATLAB

a.) $L(s) = \frac{1}{s^2+5s+20}$

$$L(j\omega) = \frac{1}{(j\omega)^2+5(j\omega)+20} = \frac{1}{(20-\omega^2)+j5\omega}$$

$$|L(j\omega)|_{dB} = 20\log 1 - \log\sqrt{(20-\omega^2)^2+(5\omega)^2}$$

$$= 20*\tfrac{1}{2}*-\log((20-\omega^2)^2+25\omega^2)$$

$$= -10\log((20-\omega^2)^2+25\omega^2)$$

$$\frac{1}{\frac{s^2}{\omega_n^2}+2\zeta\frac{s}{\omega_n}+1}$$

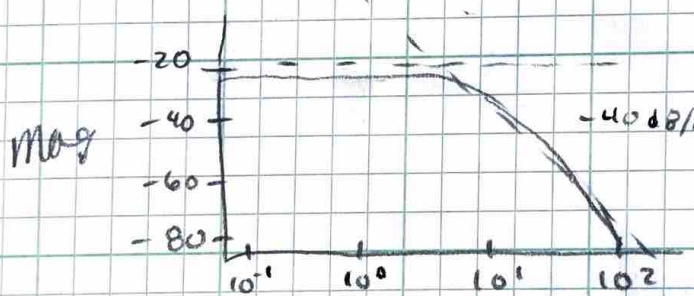
$$\hookrightarrow \frac{1/20}{\frac{s^2}{20}+\frac{s}{20}s+1}$$

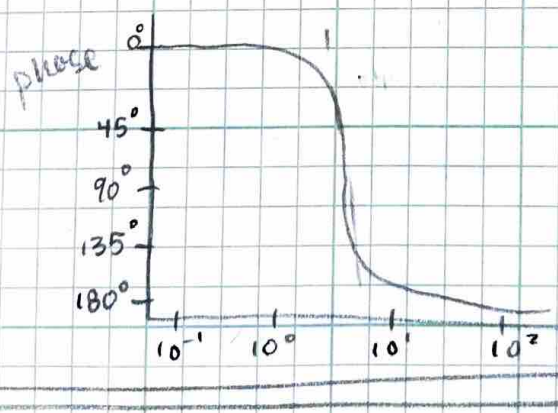
$\omega_n = \sqrt{20} = 4.47$ rad/s

$\frac{2\zeta}{\omega_n} = \frac{s}{20}$

$\zeta = \frac{s}{20}\frac{\omega_n}{2} = 0.559$

$$\begin{cases} -26.02\,dB, & \omega << 4.47 \\ -26.99, & \omega = 4.47 \\ -40\log\omega, & \omega >> 4.47 \end{cases}$$





$$\angle L(j\omega) = 0° - \angle((20-\omega^2)+j5\omega)$$

$$= \begin{cases} 0°, & \omega << 4.47 \\ -87.4°, & \omega = 4.47 \\ -180°, & \omega >> 4.47 \end{cases}$$

(6.6) Sketch asymptotes & Bode plot magnitude and phase for each open-loop TF. Verify w/ Matlab

b.) $L(s) = \frac{1}{s^3(s+8)}$

$$L(s) = \frac{1}{s^3(s+8)} = \frac{1/8}{s^3(1+s/8)}$$

$$L(j\omega) = \frac{1/8}{(j\omega)^3(1+\frac{j\omega}{8})} =$$

$$|L(j\omega)|_{dB} = 20\left(\log\tfrac{1}{8} - \log|(j\omega)^3| - \log\sqrt{1^2+(\tfrac{\omega}{8})^2}\right)$$

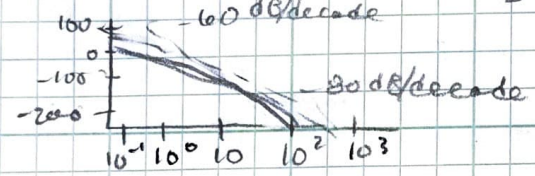
$$= 20\left(\log\tfrac{1}{8} - 3\log|j\omega| - \tfrac{1}{2}\log\left(1+\tfrac{\omega^2}{64}\right)\right)$$

$$= 20\left(\log\tfrac{1}{8} - 3\log\omega - \tfrac{1}{2}\log\left(1+\tfrac{\omega^2}{64}\right)\right)$$

$$= 20\log\tfrac{1}{8} - 60\log\omega - 10\log\left(1+\tfrac{\omega^2}{64}\right)$$

$$= \begin{cases} -18 - 60\log\omega, & \omega << 8 \\ -75.26, & \omega = 8 \\ -80\log\omega, & \omega >> 8 \end{cases}$$

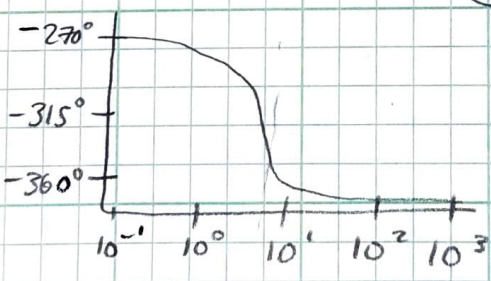
$\omega_c = 8$ rad/s



$$\angle L(j\omega) = \angle(1/8) - \angle(j\omega)^3 - \angle\left(1+\tfrac{j\omega}{8}\right)$$

$$= 0^\circ - \angle -j\omega^3 - \angle\left(1+\tfrac{j\omega}{8}\right)$$

$$= 0^\circ - 270^\circ - \begin{cases} 0^\circ, & \omega << 8 \\ 45^\circ, & \omega = 8 \\ 90^\circ, & \omega >> 8 \end{cases}$$



6.7 Sketch asymptotes of Bode plot magnitude and phase for each TF. Embellish with rough estimate for transition points. Verify w/ MATLAB

a.) $L(s) = \dfrac{s+2}{s(s+10)(s^2+2s+2)}$

$$L(s) = \frac{s+2}{s(s+10)(s^2+2s+2)} = \frac{\frac{1}{2}}{\frac{1}{2}} \rightarrow \frac{1+\frac{s}{2}}{s(s+10)\left(1+s+\frac{s^2}{2}\right)}$$

$$\frac{\frac{1}{10}}{\frac{1}{10}} \rightarrow \frac{0.1\left(1+\frac{s}{2}\right)}{s\left(1+\frac{s}{10}\right)\left(1+s+\frac{s^2}{2}\right)}$$

$$\frac{s^2}{\omega_n^2} = \frac{s^2}{2} \rightarrow \omega_n = \sqrt{2}$$

$\omega_z = 2$; $\omega_p = 10$

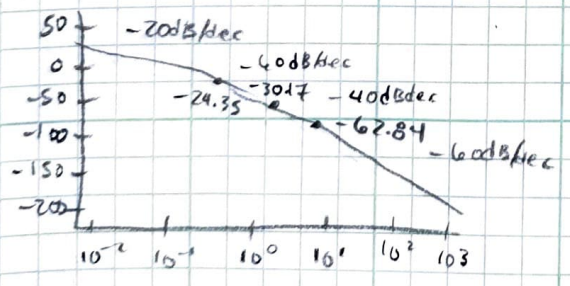
$$L(j\omega) = \frac{0.1\left(1+\frac{j\omega}{2}\right)}{(j\omega)\left(1+\frac{j\omega}{10}\right)\left(1+j\omega+\frac{(j\omega)^2}{2}\right)}$$

$$|L(j\omega)|_{dB} = 20\left(\log 0.1 + \log\left|1+\frac{j\omega}{2}\right| - \log j\omega - \log\left|1+\frac{j\omega}{10}\right| - \log\left|1+j\omega-\frac{\omega^2}{2}\right|\right)$$

$$= 20\left(-1 + \tfrac{1}{2}\log\left(1+\tfrac{\omega^2}{4}\right) - \log\omega - \tfrac{1}{2}\log\left(1+\tfrac{\omega^2}{100}\right) - \tfrac{1}{2}\log\left(\left(1-\tfrac{\omega^2}{2}\right)^2+\omega^2\right)\right)$$

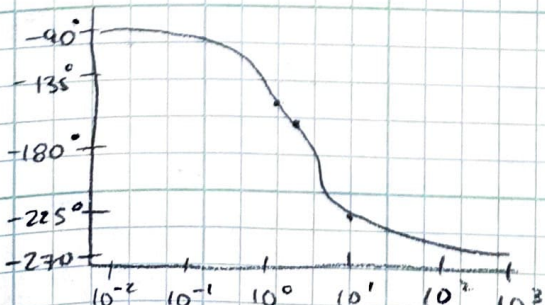
$$= -20 + 10\log\left(1+\tfrac{\omega^2}{4}\right) - 20\log\omega - 10\log\left(1+\tfrac{\omega^2}{100}\right) - 10\log\left(\left(1-\tfrac{\omega^2}{2}\right)^2+\omega^2\right)$$

$$\begin{cases} -20 - 20\log\omega & , \omega < \sqrt{2} \\ -24.35 & , \omega = \sqrt{2} \\ -13.98 - 60\log(\omega) & , \sqrt{2} < \omega < 2 \\ -30.17 & , \omega = 2 \\ -20 - 40\log\omega & , 2 < \omega < 10 \\ -62.84 & , \omega = 10 \\ -60\log\omega & , \omega > 10 \end{cases}$$



$$\angle L(j\omega) = \angle 0.1 + \angle\left(1+\frac{j\omega}{2}\right) - \angle j\omega - \angle\left(1+\frac{j\omega}{10}\right) - \angle\left(1+j\omega+\frac{(j\omega)^2}{2}\right)$$

$$\angle L(j\omega) = 0^\circ + \arctan\left(\frac{\omega}{2}\right) - 90^\circ - \arctan\left(\frac{\omega}{10}\right) - \arctan\left(\frac{\omega}{1-\frac{\omega^2}{2}}\right)$$



$$\begin{cases} -90^\circ & , \omega << \sqrt{2} \\ -152.79^\circ & , \omega = \sqrt{2} \\ -172.9^\circ & , \omega = 2 \\ -224.8 & , \omega = 10 \end{cases}$$

(6.8) Sketch the asymptotes of the Bode plot magnitude and phase for each of the listed TF. Make sure asymptotes properly take RHP singularity into account, by sketching the complex plane to see how $\angle L(s)$ changes as s goes from 0 to $+j\infty$. Verify w/ MATLAB. 5/16

a.) $L(s) = \frac{s+2}{s+10}\,\frac{1}{s^2-4}$

b.) $L(s) = \frac{s+2}{s(s+10)}\,\frac{1}{(s^2-1)}$

a.) $L(s) = \frac{s+2}{s+10}\,\frac{1}{s^2-4} = \frac{(s+2)}{s+10}\,\frac{1}{(s+2)(s-2)} = \frac{1}{(s+10)(s-2)} = \frac{-(1/10)(1/2)}{(1+\frac{s}{10})(1-\frac{s}{2})}$

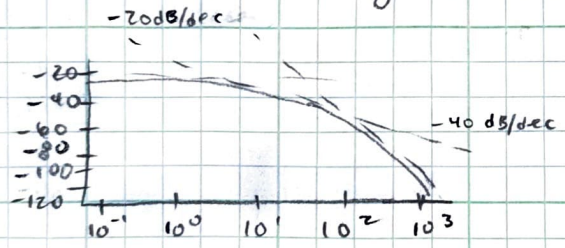
$= \frac{-0.05}{(1+\frac{s}{10})(1-\frac{s}{2})}$

$L(j\omega) = \frac{-0.05}{(1+\frac{j\omega}{10})(1-\frac{j\omega}{2})}$

$|L(j\omega)|_{dB} = 20\left(\log(|-0.05|) - \log\left|1+\tfrac{j\omega}{10}\right| - \log\left|1-\tfrac{j\omega}{2}\right|\right)$

$= 20\left(\log 0.05 - \tfrac{1}{2}\log\left(1^2+\tfrac{\omega^2}{100}\right) - \tfrac{1}{2}\left(1^2+\tfrac{\omega^2}{4}\right)\right)$

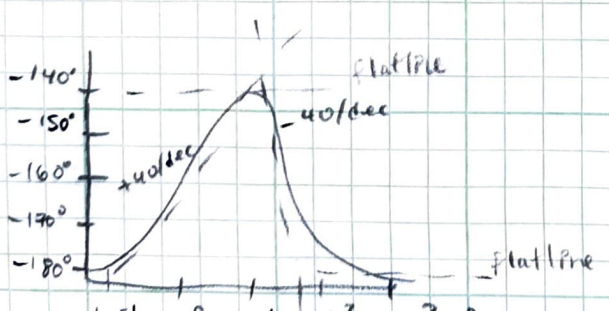
$= -26.02 - 10\log\left(1+\tfrac{\omega^2}{100}\right) - 10\log\left(1+\tfrac{\omega^2}{4}\right)$



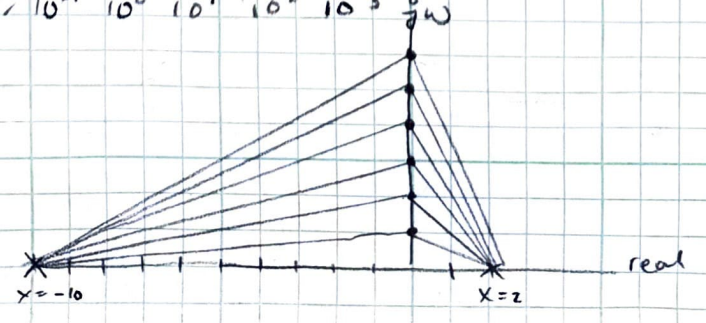
$= \begin{cases} -26.02 & ,\omega = 0 \\ -20 - 20\log\omega & \\ -29.2 & ,\omega = 2 \\ -40\log\omega & \\ -43.2 & ,\omega = 10 \end{cases}$

$\angle L(j\omega) = \angle -0.05 - \angle 1+\tfrac{j\omega}{10} - \angle 1-\tfrac{j\omega}{2}$

$= -180° - \arctan\left(\tfrac{\omega}{10}\right) - \arctan\left(\tfrac{-\omega}{2}\right)$



$= \begin{cases} 0° & ,\omega << 2 \\ -146.3° & ,\omega = 2 \\ -146.3° & ,\omega = 10 \end{cases}$



6.8 b.)

$$L(s) = \frac{s+2}{s(s+10)}\frac{1}{(s^2-1)}$$

Graph Bode plot and complex plane

$$L(s) = \frac{s+2}{s(s+10)}\frac{1}{(s^2-1)} = \frac{s+2}{s(s+10)(s+1)(s-1)}$$

$$\frac{\frac{1}{2}}{\frac{1}{2}} * \frac{\frac{1}{10}}{\frac{1}{10}} * \frac{s+2}{s(s+10)(s+1)(s-1)} = \frac{-0.1(1+\frac{s}{2})}{0.5\,s(1+\frac{s}{10})(1+s)(1-s)}$$

$$= \frac{-0.2\left(1+\frac{s}{2}\right)}{s\left(1+\frac{s}{10}\right)(1+s)(1-s)}$$

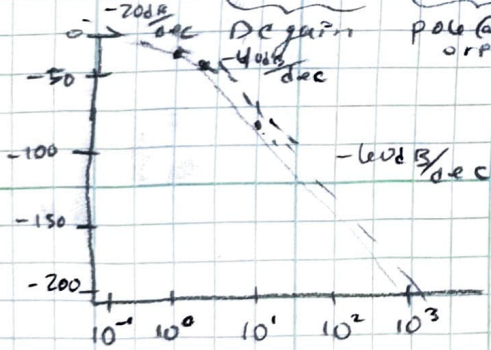
$$L(j\omega) = \frac{-0.2\left(1+\frac{j\omega}{2}\right)}{j\omega\left(1+\frac{j\omega}{10}\right)(1+j\omega)(1-j\omega)}$$

$$|L(j\omega)|_{dB} = 20\left(\log(10.21) + \tfrac{1}{2}\log\left(1+\frac{\omega^2}{4}\right) - \log\omega - \tfrac{1}{2}\log\left(1+\frac{\omega^2}{100}\right) - \tfrac{1}{2}\log(1+\omega^2) - \tfrac{1}{2}\log(1+\omega^2)\right)$$

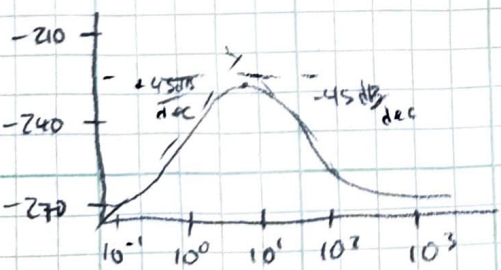
$$= -13.98 + 10\log\left(1+\frac{\omega^2}{4}\right) - 20\log\omega - 10\log\left(1+\frac{\omega^2}{100}\right) - 20\log(1+\omega^2)$$

$$= -13.98 - 20\log\omega + 20\log(1+\omega^2) + 10\log\left(1+\frac{\omega^2}{4}\right) - 10\log\left(1+\frac{\omega^2}{100}\right)$$

DC gain; pole @ origin; pole @ 1; 0 dB at 2; pole at 10



$$= \begin{cases} -13.98 - 20\log\omega, & \omega < 1 \\ -13.98, & \omega = 1 \\ -13.98 - 20\log\omega - 20\log\omega^2, & 1 < \omega < 2 \\ -30.97, & \omega = 2 \\ -13.98 - 40\log\omega, & 2 < \omega < 10 \\ -62.93, & \omega = 10 \\ -13.98 - 60\log\omega, & \omega > 10 \end{cases}$$



phase:

$$\angle L(j\omega) = \angle -0.2 + \angle 1+\frac{j\omega}{2} - \angle j\omega - \angle 1+\frac{j\omega}{10} - \angle 1+j\omega - \angle 1-j\omega$$

$$= -180^\circ + \tan^{-1}\left(\frac{\omega}{2}\right) - 90^\circ - \tan^{-1}\left(\frac{\omega}{10}\right) - \tan^{-1}(\omega) - \tan^{-1}(-\omega)$$

$$= -180^\circ - 90^\circ - \tan^{-1}(\omega) + \tan^{-1}(+\omega) + \tan^{-1}\left(\frac{\omega}{2}\right) - \tan^{-1}\left(\frac{\omega}{10}\right)$$

$$= \begin{cases} -270^\circ, & \omega < 1 \\ +45\,dB/dec, & 2 < \omega < 10 \\ -45\,dB/dec & \end{cases}$$

